

Alex Spigler

alex.spigler@gmail.com | alexspigler.dev | linkedin.com/in/alexspigler | github.com/alexspigler

EDUCATION

George Washington University

Dual Degree: BS in Statistics, BA in Computer Science; Minor in Mathematics

GPA: 3.77 | Statistics GPA: 3.88

Expected May 2027

Washington, DC

EXPERIENCE

Data Analyst Intern

AnswersNow

Jul. – Aug. 2026

Remote

- Authored a data-rules specification defining ~25 client and utilization metrics at the warehouse column and flag level, resolving a cross-team cancellation-rate discrepancy caused by tentative schedule records and standardizing executive-reporting definitions.
- Proposed and completed a manager-approved competing-risks analysis of 2,121 clients using cause-specific Cox models; a 10-percentage-point higher Stage-1 attendance rate was associated with lower dropout hazard at day 90 (HR 0.840) and day 365 (HR 0.936), with time-varying and influence diagnostics.
- Analyzed 23,406 expected client-weeks across 1,232 clients in a production PostgreSQL warehouse, integrating billing, appointment, authorization, and payer data into a 39-chart Mode report that found the weekly share receiving therapy fell from 86–89% to 78–81% while monthly reach held near 88%.

Barista

Tatte Bakery & Cafe

Feb. 2023 – Nov. 2024

Washington, DC

- Trained 7 new employees on store operations and customer service after consistently reliable performance.

Digital Asset Investor

Self-Employed

Jan. 2013 – Aug. 2023

Washington, DC

- Managed digital-asset capital for myself, family, and close associates across multiple market cycles, including the 2018 and 2022 drawdowns; shifted to passive management in 2023 to pursue formal training in Statistics and Computer Science.

PROJECTS

U.S. Unemployment Rate Forecasting | *R*

- Built a reproducible forecasting study on 942 observed monthly BLS unemployment rates (1948–2026), preserving a source-documented missing month and locking the final 24 calendar months as holdout.
- Compared five SARIMA specifications and ETS across 10 rolling validation windows; training-selected ETS cut holdout RMSE 31.4% versus naive and 43.2% versus seasonal-naive forecasts.

Chaotic Attractors: Computational Exploration & Visualization | *Python*

- Built a pip-installable Python library (CLI + API) for searching, validating, and visualizing 9 four-parameter 2D systems (5 established forms, 4 custom variants).
- Designed a seeded, staged search that removes burn-in and validates survivors with finite-time largest Lyapunov estimates; ranked outputs by geometry and rendered density with subsampled Gaussian KDE.

TECHNICAL SKILLS

Languages: Python, R, SQL, Java

Statistical Methods: Experimental Design & Power Analysis, Bayesian Inference & MCMC, Regression Diagnostics, Survey Sampling & Weighting, Time Series & Intervention Analysis, Survival Analysis & Competing Risks

Mathematical Methods: Linear Algebra, Dynamical Systems & Stability Analysis, Numerical ODE Methods

Libraries & Tools: NumPy, pandas, SciPy, Matplotlib, Git, pytest, Mode Analytics, R Markdown

Databases: PostgreSQL